

Lindacin Capsules 150 mg "S.T."(Clindamycin)

[INGREDIENT]

Each capsule contains:

Clindamycin (as Hydrochloride).....150mg

[CLINICAL PHARMACOLOGY]

PHARMACOLOGICAL ACTION

Clindamycin hydrochloride binds exclusively to the 50S subunit of bacterial ribosomes and suppresses protein synthesis. Clindamycin has antibacterial activity against gram-positive organisms and a lower order of activity against gram-negative organisms. In vitro activity does not necessarily imply in vivo efficacy. The in vitro spectrum of activity includes staphylococci (including penicillinase-producing strains), beta-haemolytic streptococci, *Diplococcus pneumoniae*, *Clostridium perfringens*, *Corynebacterium acnes* and *Actinomyces israelii*. Clindamycin hydrochloride is not active against most strains of *Streptococcus faecalis*, *Escherichia coli*, *Shigella* spp., *Salmonella* spp., *Proteus* spp., and *Pseudomonas* spp.

Clindamycin is rapidly absorbed after oral administration (peak blood levels occurred in 45 minutes). Bone and other body fluid levels are obtained rapidly. Absorption is almost complete (90%). Blood levels exceed the minimum inhibitory concentration (MIC) for most indicated organisms for at least six hours following administration of the usually recommended doses.

The biological half-life is 2,4 hours.

PHARMACOKINETICS

Microbiology

Clindamycin has been shown to have in vitro activity against isolates of the following organisms:

Aerobic gram-positive cocci, including:

Staphylococcus aureus,

Staphylococcus epidermidis, (penicillinase and nonpenicillinase producing strains). When tested by in vitro methods some staphylococcal strains originally resistant to erythromycin rapidly develop resistance to clindamycin,

Streptococci (except *Streptococcus faecalis*),

Pneumococci.

Anaerobic gram-negative bacilli, including:

Bacteroides species (including *Bacteroides fragilis* group and *Bacteroides melaninogenicus* group), *Fusobacterium* species.

Anaerobic gram-positive nonsporeforming bacilli, including:

Propionibacterium,

Eubacterium,

Actinomyces species.

Anaerobic and microaerophilic gram-positive cocci, including:

Peptococcus species,

Peptostreptococcus species,

Microaerophilic streptococci.

Clostridia: Clostridia are more resistant than most anaerobes to clindamycin. Most *Clostridium perfringens* are susceptible, but other species, e.g., *Clostridium sporogenes* and *Clostridium tertium* are frequently resistant to clindamycin. Susceptibility testing should be done.

Cross resistance has been demonstrated between clindamycin and lincomycin.

Antagonism has been demonstrated between clindamycin and erythromycin.

Human pharmacology

Serum level studies with a 150 mg oral dose of clindamycin hydrochloride in 24 normal adult volunteers showed that clindamycin was rapidly absorbed after oral administration. An average peak serum level of 2.50 mcg/mL was reached in 45 minutes; serum levels averaged 1.51 mcg/mL at 3 hours and 0.70 mcg/mL at 6 hours. Absorption of an oral dose is virtually complete (90%), and the concomitant administration of food does not appreciably modify the serum concentrations; serum levels have been uniform and predictable from person to person and dose to dose. Serum level studies following multiple doses of clindamycin hydrochloride for up to 14 days show no evidence of accumulation or altered metabolism of drug.

Serum half-life of clindamycin is increased slightly in patients with markedly reduced renal function. Hemodialysis and peritoneal dialysis are not effective in removing clindamycin from the serum.

Concentrations of clindamycin in the serum increased linearly with increased dose. Serum levels exceed the MIC (minimum inhibitory concentration) for most indicated organisms for at least six hours following administration of the usually recommended doses.

Clindamycin is widely distributed in body fluids and tissues (including bones). The average biological half-life is 2.4 hours.

Approximately 10% of the bioactivity is excreted in the urine and 3.6% in the feces; the remainder is excreted as bioinactive metabolites.

Doses of up to 2 grams of clindamycin per day for 14 days have been well tolerated by healthy volunteers, except that the incidence of gastrointestinal side effects is greater with the higher doses.

No significant levels of clindamycin are attained in the cerebro-spinal fluid, even in the presence of inflamed meninges.

Pharmacokinetic studies in elderly volunteers (61-79 years) and younger adults (18-39 years) indicate that age alone does not alter clindamycin pharmacokinetics (clearance, elimination half-life, volume of distribution, and area under the serum concentration-time curve) after IV administration of clindamycin phosphate. After oral administration of clindamycin hydrochloride, elimination half-life is increased to approximately 4.0 hours (range 3.4-5.1 h) in the elderly compared to 3.2 hours (range 2.1 - 4.2 h) in younger adults. The extent of absorption, however, is not different between age groups and no dosage alteration is necessary for the elderly with normal hepatic function and normal (age-adjusted) renal function.

[INDICATIONS]

Clindamycin is indicated in the treatment of serious infections caused by susceptible anaerobic bacteria.

Clindamycin is also indicated in the treatment of serious infections due to susceptible strains of streptococci, pneumococci, and staphylococci. Its use should be reserved for penicillin allergic patients or other patients for whom, in the judgment of the physician, a penicillin is inappropriate. Because of the risk of colitis, as described in the Description

Anaerobes: Serious respiratory tract infections such as emphysema, anaerobic pneumonitis and lung abscess; serious skin and soft tissue infections; septicemia; intra-abdominal infections such as peritonitis and intra-abdominal abscess (typically resulting from anaerobic organisms resident in the normal gastrointestinal tract); infections of the female pelvis and genital tract such as endometriosis, nongonococcal tubo-ovarian abscess, pelvic cellulitis and postsurgical vaginal cuff infection.

Streptococci: Serious respiratory tract infections; serious skin and soft tissue infections.

Staphylococci: Serious respiratory tract infections; serious skin and soft tissue infections.

Pneumococci: Serious respiratory tract infections.

Bacteriologic studies should be performed to determine the causative organisms and their susceptibility to clindamycin.

[DOSAGE AND ADMINISTRATION]

Oral

Adults:

Mild to moderately severe infections - 150 to 300 mg every 6 hours.

Severe infections - 300 to 450 mg every 6 hours.

Children:

Mild to moderately severe infections - 8 to 16 mg/kg/day (4 to 8 mg/lb/day) divided into three or four equal doses.

Severe infections - 16 to 20 mg/kg/day (8 to 10 mg/lb/day) divided into three or four equal doses.

To avoid the possibility of oesophageal irritation, clindamycin hydrochloride capsules should be taken with a full glass of water and no less than 30 minutes before lying down.

[INTERACTIONS WITH OTHER MEDICAMENTS]

Clindamycin has been shown to have neuromuscular blocking properties that may enhance the action of other neuromuscular blocking agents. Therefore, it should be used with caution in patients receiving such agents. Antagonism has been demonstrated between clindamycin and erythromycin in vitro.

Because of possible clinical significance, these two drugs should not be administered concurrently.

[PREGNANCY AND LACTATION]

Pregnancy: Teratogenic effects

Pregnancy category B

Reproduction studies performed in rats and mice using oral doses of clindamycin up to 600 mg/kg/day (3.2 and 1.6 times the highest recommended adult human dose based on mg/m², respectively) or subcutaneous doses of clindamycin up to 250 mg/kg/day (1.3 and 0.7 times the highest recommended adult human dose based on mg/m², respectively) revealed no evidence of teratogenicity.

There are, however, no adequate and well-controlled studies in pregnant women. Because animal reproduction studies are not always predictive of the human response, this drug should be used during pregnancy only if clearly needed.

Nursing Mothers

Clindamycin has been reported to appear in breast milk in the range of 0.7 to 3.8 mcg/mL.

[EFFECTS ON ABILITY TO DRIVE & USE MACHINE]

Clindamycin has no or negligible influence on the ability to drive and use machines.

[SIDE EFFECTS]

The following reactions have been reported with the use of clindamycin.

Gastrointestinal: Abdominal pain, esophagitis, nausea, vomiting and diarrhea.

Hypersensitivity Reactions: Generalized mild to moderate morbilliform-like (maculopapular) skin rashes are the most frequently reported adverse reactions. Vesicubullous rashes, as well as urticaria, have been observed during drug therapy. Rare instances of erythema multiforme, some resembling Stevens-Johnson syndrome, and a few cases of anaphylactoid reactions have also been reported.

Skin and Mucous Membranes: Pruritus, vaginitis, and rare instances of exfoliative dermatitis have been reported.

Liver: Jaundice and abnormalities in liver function tests have been observed during clindamycin therapy.

Renal and urinary disorders: Acute kidney injury (the frequency is not known)

Hematopoietic: Transient neutropenia (leukopenia) and eosinophilia have been reported. Reports of agranulocytosis and thrombocytopenia have been made. No direct etiologic relationship to concurrent clindamycin therapy could be made in any of the foregoing.

Gastrointestinal disorders

Frequency 'not known': Oesophageal ulcer, oesophagitis

[WARNINGS]

Clindamycin therapy has been associated with severe colitis which may end fatally.

It should be reserved for serious infections where less toxic antimicrobial agents are inappropriate.

It should not be used in patients with nonbacterial infections, such as most upper respiratory tract infections.

Its use in newborns is contraindicated.

Clindamycin is potentially nephrotoxic. Acute kidney injury including acute renal failure has been reported. Therefore, monitoring of renal function should be considered during therapy of patients with pre-existing renal dysfunction or taking concomitant nephrotoxic drugs and monitoring of renal function should be performed if therapy is prolonged.

A careful inquiry should be made concerning previous sensitivities to drugs and other allergens.

Usage in Pregnancy - Safety for use in pregnancy has not been established.

Usage in Newborns and Infants - When clindamycin hydrochloride is administered to newborns and infants, appropriate monitoring of organ system functions is desirable.

Nursing Mothers - Clindamycin has been reported to appear in breast milk in ranges of 0.7 to 3.8 mcg/mL.

Usage in Meningitis - Since clindamycin does not diffuse adequately into the cerebrospinal fluid, the drug should not be used in the treatment of meningitis.

Antagonism has been demonstrated between clindamycin and erythromycin in vitro. Because of possible clinical significance, these two drugs should not be administered concurrently.

[PRECAUTIONS]

Review of experience to date suggests that a subgroup of older patients with associated severe illness may tolerate diarrhea less well.

When clindamycin is indicated in these patients, they should be carefully monitored for change in bowel frequency.

Clindamycin hydrochloride should be prescribed with caution in individuals with a history of gastrointestinal disease, particularly colitis.

Clindamycin hydrochloride should be prescribed with caution in atopic individuals.

During prolonged therapy, periodic liver and kidney function tests and blood counts should be performed.

Indicated surgical procedures should be performed in conjunction with antibiotic therapy.

The use of clindamycin hydrochloride occasionally results in over growth of nonsusceptible organisms - particularly yeasts. Should superinfections occur, appropriate measures should be taken as indicated by the clinical situation.

Patients with very severe renal disease and/or very severe hepatic disease accompanied by severe metabolic aberrations should be dosed with caution, and serum clindamycin levels monitored during high-dose therapy.

Clindamycin has been shown to have neuromuscular blocking properties that may enhance the action of other neuromuscular blocking agents. Therefore, it should be used with caution in patients receiving such agents.

Due to the risk of oesophagitis and oesophageal ulcer, it is important to ensure compliance with administration guidance (see Sections Route of administration and Side Effects).

[CONTRAINDICATIONS]

Clindamycin hydrochloride is contraindicated in individuals with a history of hypersensitivity to preparations containing clindamycin or lincomycin.

[SYMPTOMS AND TREATMENT OF OVERDOSE]

Significant mortality was observed in mice at an intravenous dose of 855 mg/kg and in rats at an oral or subcutaneous dose of approximately 2618 mg/kg. In the mice, convulsions and depression were observed. Hemodialysis and peritoneal dialysis are not effective in removing clindamycin from the serum.

[PRODUCT DESCRIPTION]

It is white powder, filled in the No.1 capsule which is red cap and purple body printed “ST₀₂₃” on the cap and body in white ink.

[STORAGE CONDITION]

Store at controlled room temperature to 30° C.

[SHELF LIFE]

Two years

[PACKAGE] 10 Capsules in PVC/foil blister strip, 10 strips in an aluminum bag / box

REGISTRATION NO.: MAL07082868AZ

Manufacturer:

TAIWAN BIOTECH CO., LTD.

22, Chieh Shou Road, Taoyuan, Taiwan, R.O.C.

Product Registration Holder:

SUNWARD PHARMACEUTICAL SDN BHD

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15 May 2026